PROBLEM–SOLUTION FIT

FitBuddy – AI Fitness Plan Generator using Gemini Models

Date	01 October 2026	**Team ID**	SWTID-2026-2430
Project Name	Team Enrollment Confirmation – FitBuddy – AI Fitness Plan Generator using Gemini Models	**Maximum Marks**	5 Marks
Team Members	Divya.D, Deepika.M, Dharani.T, Divya shree.R, DivyaDharshini.V	**Technology**	FastAPI, Gemini models, HTML/Jinja2, SQLite

Problem-Solution Fit Canvas

The canvas below is tailored to FitBuddy's stated purpose: personalized 7-day workout plans, feedback-based updates, nutrition/recovery tips, and an admin view.

1. CUSTOMER SEGMENTS [CS] People who want a structured fitness routine; beginners and general wellness users; people pursuing goals such as strength, flexibility, or weight management; coaches/admins overseeing multiple users.	6. CUSTOMER LIMITATIONS [CL] Limited time or experience in planning workouts; differing fitness levels and equipment access; limited access to a personal coach; internet/API availability and privacy considerations.	5. AVAILABLE SOLUTIONS — PROS & CONS [AS] General online workout plans, fitness apps, trainer advice, and manual planning. Pros: easy to find and familiar. Cons: may not match each user's goal, intensity, or feedback as directly.
2. PROBLEMS / PAINS + FREQUENCY [PR] Users may be unsure what exercises to do, how to organize a weekly plan, or how to adapt a routine when preferences change. These needs can arise whenever they start or revise a fitness routine.	9. PROBLEM ROOT / CAUSE [RC] Workout planning requires users to combine goals, intensity, weekly structure, and ongoing adjustments. Generic plans and manual planning can be difficult to personalize and maintain.	7. BEHAVIOR + INTENSITY [BE] Users enter basic details and goals, review the generated 7-day plan, request changes through feedback, and check nutrition/recovery tips. Engagement may vary with time, confidence, and consistency.
3. TRIGGERS TO ACT [TR] Starting a fitness goal; wanting a more organized weekly schedule; feeling a current plan is too difficult or repetitive; wanting a nutrition or recovery suggestion.	10. YOUR SOLUTION [SL] FitBuddy uses Gemini models through FastAPI to generate a personalized 7-day workout plan, provide a concise nutrition/recovery tip, and revise the plan from user feedback. SQLite stores user details and original/updated plans; an admin view supports oversight.	8. CHANNELS OF BEHAVIOR [CH] **Online:** FitBuddy web interface, form submission, generated results, feedback form. **Offline:** perform the suggested activities, follow recovery guidance, and discuss plans with a coach if needed.
4. EMOTIONS BEFORE / AFTER [EM] Before: uncertain, overwhelmed, or unmotivated by planning. After: more organized and supported by a clear plan and a way to request adjustments.	**FITBUDDY VALUE** A simple, feedback-aware planning flow that keeps workout structure, practical tips, and stored plan history in one web application.	**USER JOURNEY** Enter details → receive plan and tip → submit feedback if needed → review updated plan.

TEAM WORK ALLOCATION

Responsibilities assigned in the same order as the team-member list provided.

1	**Divya.D**	Project coordination and requirements	Define customer problem, target users, fitness goals/intensity inputs, and overall workflow; consolidate team work and documentation.
2	**Deepika.M**	Gemini AI integration and prompt design	Work on workout-plan generation, feedback-based plan updates, and prompts for structured 7-day output using Gemini models.
3	**Dharani.T**	FastAPI backend and routing	Handle form processing and routes for plan generation and feedback; connect backend routes to AI functions.
4	**Divya shree.R**	Frontend and user experience	Build/maintain HTML and Jinja2 pages for user input, results, nutrition/recovery tip, and feedback; support responsive styling.
5	**DivyaDharshini.V**	Database, testing, and deployment support	Support SQLite/SQLAlchemy storage of users and original/updated plans; test main flows and verify local deployment and API docs.

Shared team responsibility

All members contribute to reviewing the final output, checking that generated plans are clearly presented, and preparing the project demonstration. Fitness suggestions should be treated as general information and adapted to individual needs.

Project technology: FastAPI • Google Gemini models • HTML/CSS/Jinja2 • SQLite/SQLAlchemy • Uvicorn